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Porosity prediction in laser beam welding with a multimodal physics-informed machine learning framework

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ABSTRACT

Laser beam welding (LBW) of metallic components is a knowledge-intensive manufacturing process whose quality depends on the complex multi-physics. However, its engineering application is often hindered by the occurrence of porosity defects. Achieving a thorough understanding and reliable prediction of porosity defects remains difficult because it demands robust representation and reasoning over nonlinear and hard-to-observe physical information. In this study, we propose an integrated multimodal physics-informed machine learning (PIML) framework with the help of multi-physical modelling and experimental data to predict the porosity defects in laser beam welding of aluminum alloys. The whole framework contains a multimodal PIML model for predicting the porosity ratio and an ML-based estimator for relevant physical information. By utilizing the scalar welding parameters and high-dimensional physical information (probability of keyhole collapses, cumulative existing time of collapses, and molten pool geometry) as inputs, the multimodal PIML model shows great superiority in predicting the porosity ratio, with a reduction of the mean square error by 45%, compared with the ML model trained only with welding parameters. The ML-based estimator constructed with an encoder-decoder architecture can accurately reproduce the critical physical information within a timeframe of seconds. By integrating these two ML models, the proposed framework advances engineering informatics by offering a scalable, physics-knowledge-centric solution for fast and accurate porosity prediction in LBW manufacturing.

1. Introduction

In modern manufacturing industries, laser beam welding (LBW) operating in keyhole mode is increasingly recognized as an advanced method for metal joining [1]. This technique offers distinct advantages over traditional arc welding methods, such as reduced energy input, enhanced penetration depth, increased welding speed, a narrower heat-affected zone, and reduced distortion in welded components. Despite these benefits, porosity defects remain one of the most critical issues in LBW across various metallic materials, including lightweight Al- and Mg-alloys, known for their low density and high thermal conductivity, as well as steels, Ti-alloys, and Ni-alloys [2,3]. Porosity significantly compromises the structural continuity and uniformity of weld properties, creating stress concentrations that may result in premature structural failures. Therefore, accurate prediction of porosity formation is crucial for achieving automated and intelligent LBW manufacturing.

Process porosity defects are closely related to bubble formation resulting from fluctuations or collapses of the keyhole, followed by bubble migration through the molten pool [4–6]. Unfortunately, its prediction remains challenging due to the complex, multi-coupled, and non-linear physics involved, as well as the inherent difficulties of experimental detection within opaque metallic materials [7,8]. Parametric studies using statistical Design of Experiment (DoE) methods provide a practical approach to process mapping, effectively addressing the high-dimensional interactions between welding parameters [3,9]. However, results from these methods typically show limited generalizability and transferability across broader process conditions.

To address these limitations, online detection techniques have been developed to establish relationships between porosity formation and the physical features of the molten pool, such as data sources from high-speed imaging, infrared (IR) thermal imaging, and acoustic emissions [10–12]. These datasets are, however, often multi-dimensional in both the time and frequency domains, e.g., a time-varying series of temperature distribution from the IR images, which leads to difficulties in

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correlating with the porosity formation. Feature selections were, therefore, often employed manually to extract one or two key characteristics from the measured data. For example, the peak intensity in IR images was selected to align with porosity distribution data obtained from μ-CT measurements [11]. In contrast, machine learning (ML) techniques offer the potential to integrate online detection data directly, leveraging visual and thermal data from the keyhole or molten pool for robust porosity prediction. In the study by Zhang et al. [13], in-process molten pool images were captured using a coaxial high-speed camera and then labeled with ex-situ measured porosity attributes. These images served as input for a convolutional neural network (CNN) model designed to identify porosity. Similarly, Ma et al. employed signal spectrum graphs derived from keyhole opening images, processed through wavelet packet decomposition, to train a CNN model for detecting porosity locations [14]. However, multiple studies have shown that ML models exhibit lower accuracy, typically a 10% to 20% reduction, when predicting deeper pores [13,15]. This decline in performance is primarily due to the absence of thermo-fluid data within the molten pool during model training. To enhance the model's performance, some researchers have integrated the 1D keyhole depth from Optical Coherence Tomography (OCT) [16–18] and 2D keyhole profiles obtained through synchrotron X-ray imaging [15] into the dataset.

Over the past five years, a novel approach known as physics-informed machine learning (PIML) has gained significant attention for its superior performance in predicting complex physical phenomena, such as the defect formation in welding and additive manufacturing [19,20]. The training data are acquired from physical models governed by essential conservation equations so that the fundamental physical laws are also implicitly considered. A well-calibrated numerical model is often employed as a reliable alternative for generating extensive and quantitative molten pool data for training. Compared to experimental measurements, numerical models can provide a more detailed spatial and temporal distribution of key variables inside the molten pool, such as the transient 3D evolution of keyholes or the solidification rate at critical locations. In the work of Gawade et al. [21], a combination of a thermal finite-element model and a CNN framework was used to predict the occurrence of keyhole-induced porosity and pore size. This was achieved by selecting physical information, such as the molten pool dimensions and peak temperature, along with infrared images as inputs. A deep belief network trained on a fused dataset, which included the numerically calculated maximum temperature on the keyhole wall and the OCT-measured keyhole depth, was employed to predict porosity levels in LBW of aluminum [22]. Recently, the authors proposed a PIML framework by utilizing multi-physical modeling and experimental data to predict the porosity ratio during LBW. With a comprehensive selection of features concerning solidification, liquid metal flow, keyhole stability, and molten pool geometry, the PIML model shows a higher accuracy in predicting the porosity ratio. Furthermore, the hierarchical importance of the physical factors on the porosity formation was revealed based on the well-trained PIML model [23].

Although the PIML method has been successfully applied in predicting complex phenomena with highly non-linear and multi-coupled physics, several noticeable limitations must be mentioned. First, the physical information was only selected and integrated as the scalar data in the existing studies (molten pool length, maximum temperature, or keyhole collapse frequency, etc.), and the advantages of the multi-physical models, which can describe 2D or 3D physical fields, have not been fully utilized yet [20–23]. Secondly, easy access to physical information in a realistic time frame is still difficult due to the computational intensity of the numerical models (tens of hours for a single simulation case).

Building on our previous PIML framework [23], this paper aims to develop a multimodal PIML model that integrates high-dimensional physical data generated by a multi-physics model along with welding parameters (as scalar data), achieving an accurate and robust prediction of the porosity formation in LBW. In contrast to our earlier work, in which all physical information was reduced to time-averaged scalar features or dimensionless groups, the present study introduces a fundamentally different representation and learning strategy by directly incorporating spatially resolved, high-dimensional physical information into the ML framework, such as the probability distribution of keyhole collapse and the variation of molten pool length along the thickness direction. To enhance the usability of the PIML model, an additional ML model with an encoder-decoder architecture is trained using welding parameters as inputs for a fast estimation of the selected high-dimensional physical data (the inputs of the multimodal PIML) within a timeframe of seconds. By integrating the multimodal PIML model with the fast physical-feature estimator, the resulting framework retains the physics-based interpretability of the PIML approach while overcoming the prohibitive computational cost of direct multi-physics simulations. A detailed comparison between the previous PIML framework and the present work is given in Table 1.

Table 1
Comparison between the previous PIML framework [23] and the present work.

Aspect	Ref. [23]	Present work
Experimental data	Same LBW dataset	Same LBW dataset
Multi-physics model	Same validated model	Same validated model
Physical feature form	Scalar, time-averaged	Spatially resolved 1D fields
Keyhole description	Depth, collapse frequency	Collapse probability + duration distributions
Molten pool description	Single characteristic length	Thickness-dependent profile
ML architecture	Fully connected DNN	Multimodal CNN–DNN
Physics integration	Feature-level	Architecture-level
Surrogate model	Not implemented	Encoder–decoder physical estimator
Prediction time	Hours to days	Seconds
Main contribution	Physics insight & hierarchy	Deployable integrated framework

2. Methodology of raw data acquisition

2.1. LBW experiments and characterization

With a Trumpf disk laser system, the LBW experiments were conducted in a bead-on-plate mode on AlMg3 alloy plates, measuring 300 mm × 90 mm × 10 mm. The laser (1030 nm wavelength) had a forward angle of 10 degrees to the normal direction of the base metal surface. Before welding, mechanical cleaning was performed to remove the oxidation layer and minimize the impact of metallurgical porosity. A customized multi-level Latin hypercube sampling (LHS) method was implemented to generate a design matrix for the construction of the dataset, incorporating welding parameters of laser power (4 kW–8 kW), welding speed (2 m/min–5 m/min), focus position (0 mm – −6 mm), and laser diameter (0.5 mm–0.8 mm). The complete matrix is available in the supplementary materials. The discrepancy, which is a classical and widely recognized metric for evaluating uniformity, of 0.09 indicated a satisfactory level of uniformity. Regarding coverage, all marginal levels of each factor were fully represented (100% marginal coverage). The pairwise coverage ranged between 75% and 100%, ensuring that most two-factor interactions were captured. In total, 66 combinations were obtained out of 320 possible, corresponding to a full factorial coverage of 20.6%, which is typical for Latin Hypercube or fractional factorial designs. Because of rounding applied to discretized parameter levels in the multi-level LHS design, a small number of duplicated parameter combinations occurred in the design matrix, resulting in 63 unique parameter combinations. In the present work, these duplicates were retained and treated as independent samples, reflecting minor variations in the experimentally observed outputs for identical input conditions.

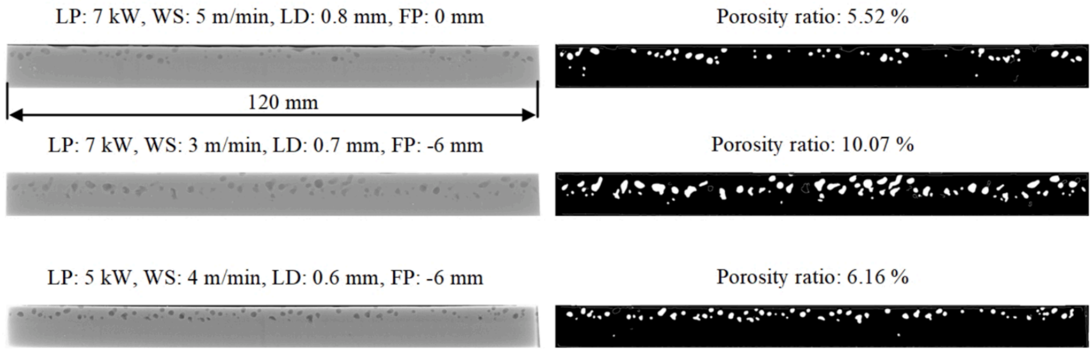

Fig. 1. Representative X-ray images and the processed images for porosity recognition.

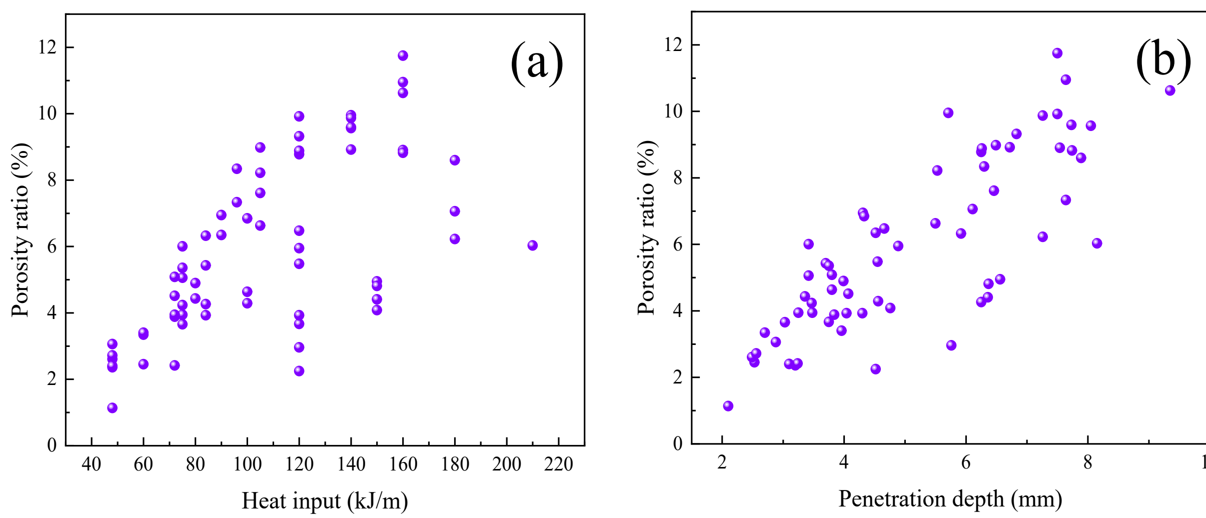

Fig. 2. The correlation of porosity ratio with (a) heat input and (b) penetration depth.

A section of 120 mm × 10 mm × 10 mm was mechanically cut from each weld for X-ray analysis to determine the porosity ratio, which served as the dataset label. The specimen was extracted from the central region along the welding direction to ensure a quasi-steady-state welding condition and to avoid edge effects near the start and end of the weld. The X-ray beam was directed perpendicularly to the welding direction, capturing the porosity distribution in the longitudinal section. The imaging parameters were set to a voltage of 50 kV, a tube current of 2.4 mA, a focal spot size of 3 mm, a film-to-focus distance of 1000 mm, and an exposure time of 16.7 min. An in-house developed recognition algorithm was used to automatically identify the porosity contour in the X-ray images. The porosity ratio is defined as the two-dimensional pore area fraction in the longitudinal section. During algorithm development, a set of representative X-ray images was selected and manually analyzed using ImageJ software to serve as reference data. The recognition algorithm's hyperparameter was then optimized to ensure that the algorithm-derived porosity ratios closely matched the manually measured values. Since the variability between repeated parameter combinations is consistently low, with absolute differences below 1%, it is considered negligible for the present study and is therefore not discussed further. Additional statistics for repeated parameter combinations are provided in the Supplementary Materials.

Some representative X-ray images and the corresponding processed images for the porosity recognition are shown in Fig. 1. It can be found that the porosity defect is dominated by the process pores, which are characterized by their size of more than 100 μm in diameter and irregular shapes. The correlation of the porosity ratio with heat input and penetration depth is plotted in Fig. 2. The porosity ratio shows a scattered distribution with increasing heat input and a rough positive correlation with the penetration depth. It implies the non-linear physics involved in porosity formation and the inherent difficulties in accurate prediction using only welding parameters or a single index.

2.2. Multi-physics modeling

The acquisition of the multi-dimensional physical information, as part of the inputs of the PIML model, is realized by a numerical model developed in our previous works [24,25]. The mass, momentum, and energy conservation equations coupled with the volume-of-fraction (VOF) equation are solved by ANSYS Fluent 23.1 to calculate the heat transfer, fluid flow, and free surface deformation during LBW.

A ray-tracing algorithm is employed to calculate the laser beam path, including the beam propagation before reaching the keyhole surface and subsequent multiple reflections between the keyhole walls. Each sub-ray exhibits either a focusing or defocusing feature, depending on the location of the first reflection relative to the focal position, thereby geometrically reproducing the beam caustics. To reconstruct the keyhole profile with an explicit mathematical expression, a newly developed localized Level-Set (LS) method is implemented [26]. The exact reflection points, determined by the intersection of sub-ray vectors with the LS-reconstructed free surface, can be identified accurately [27,28]. A physics-based Fresnel reflection model is applied for laser absorption calculations. This model defines absorptivity as a function of the incident angle, the laser wavelength, the surface temperature, and the material properties, which enables a reliable application in a wide range of welding parameters without the adaptation of empirical

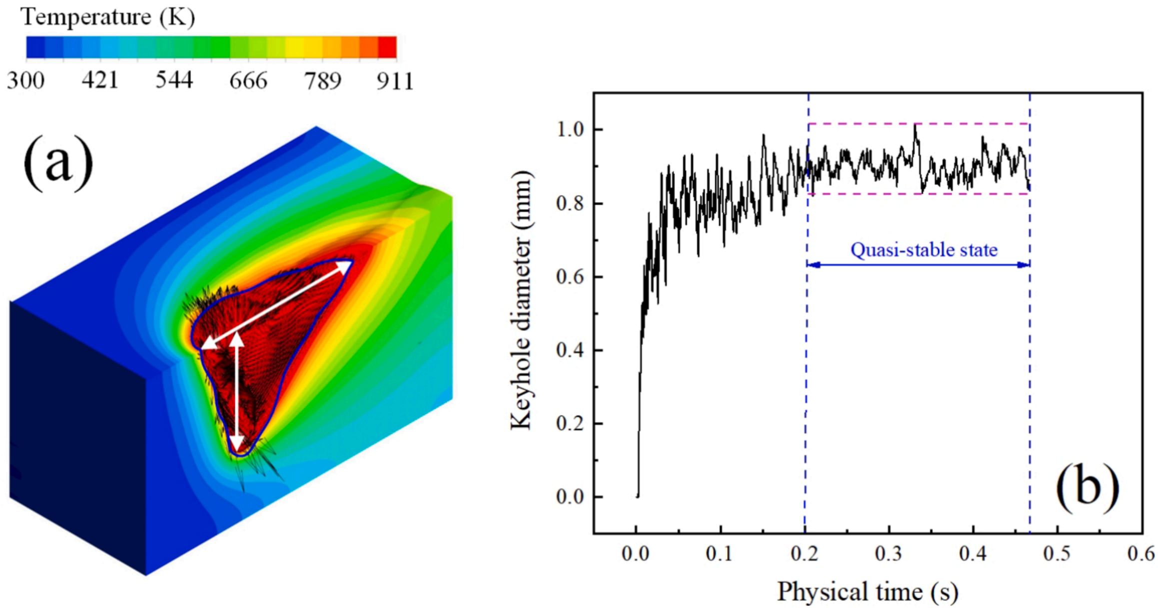

Fig. 3. Representative calculation results: (a) temperature field and fluid flow, (b) time-varying keyhole diameter.

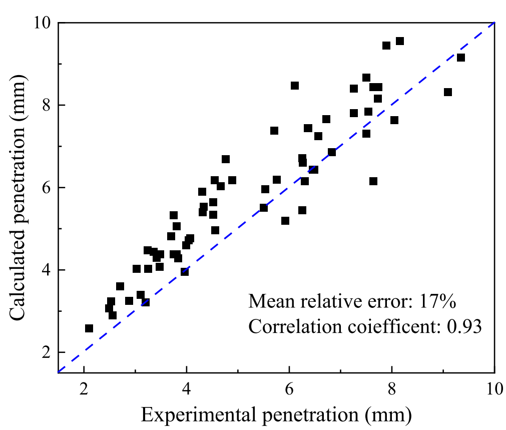

Fig. 4. Comparison of calculated and experimental penetration depth across the LHS space.

parameters [29]. In addition, other important physical factors, including the temperature-dependent recoil pressure, Marangoni force, Laplace pressure, gravity, buoyancy, etc., are also considered in this model. However, to reduce the computational cost to an affordable level, the whole formation procedure of the porosity defects, which requires extra physical models to describe the bubble formation, movement, and potential merging in the molten pool, as well as a more sophisticated solidification model to consider the capture of the bubble on the solidification front, is not simulated by the multi-physical model. As the multimodal PIML model does not explicitly utilize porosity formation features as input variables but is instead trained on welding parameters and other physical information, such as weld pool length, the exclusion of the detailed porosity formation process is not expected to influence the present ML study.

All welding parameter combinations in the LHS matrix were simulated by the multi-physics model described above. Repeated simulations were not carried out for duplicated experimental parameters, since the numerical model is deterministic and yields identical results for identical input parameters. In total, 63 simulation cases were generated to provide physically meaningful information. A physical time of 100 ms to 150 ms can be simulated every 24 h using 88 cores and 256 GB of memory in the high-performance cluster at Bundesanstalt für Materialforschung und –prüfung (BAM).

The transient 3D temperature field and fluid flow in the molten pool and the dynamic keyhole revolution can be calculated by the multi-physics model, as shown in Fig. 3(a). These results are analogous to the experimental measurements from high-speed imaging or X-ray transmission but have finer spatial resolution. Not only the characteristic dimensions (the white arrow for weld length and keyhole depth) but also 2D/3D geometries (the blue contour) can be easily extracted and analyzed. Moreover, based on the temporally and spatially well-structured calculation results, the transient evolution of critical physical information, for instance, the time-varying keyhole diameter in Fig. 3(b), can also be obtained.

The multi-physics model was validated against experimental penetration depth over the same LHS parameter space used in this study. Across all LHS points, the predicted penetration depth shows a mean relative error of 17% and a correlation coefficient of 0.93 (see Fig. 4), indicating good predictive consistency over the investigated domain. Experimental validation of molten pool length is available only for extreme parameters of the LHS space, as reported in our previous work [23].

While residual simulation errors may exist and could bias the learned feature–porosity relationships in certain regimes (e.g., low penetration/low porosity), the proposed PIML framework primarily leverages relative trends and statistical/spatial features rather than absolute physical thresholds. Uncertainty in simulation outputs is not explicitly propagated; therefore, the results of the following SHAP/PFI analysis are interpreted as qualitative indicators of feature relevance within a simulation-consistent feature space.

3. Machine learning frameworks

In this section, two ML models will be constructed and then integrated into a unified framework for a more accurate prediction of the porosity level in LBW of aluminum within an industrially applicable time frame. First, a multimodal PIML model will be developed for porosity ratio prediction by utilizing the high-dimensional physical information calculated by the numerical model, along with the welding parameters as scalar data. Subsequently, an additional ML model with an encoder-decoder architecture will be trained with the welding parameters to estimate the critical physical information.

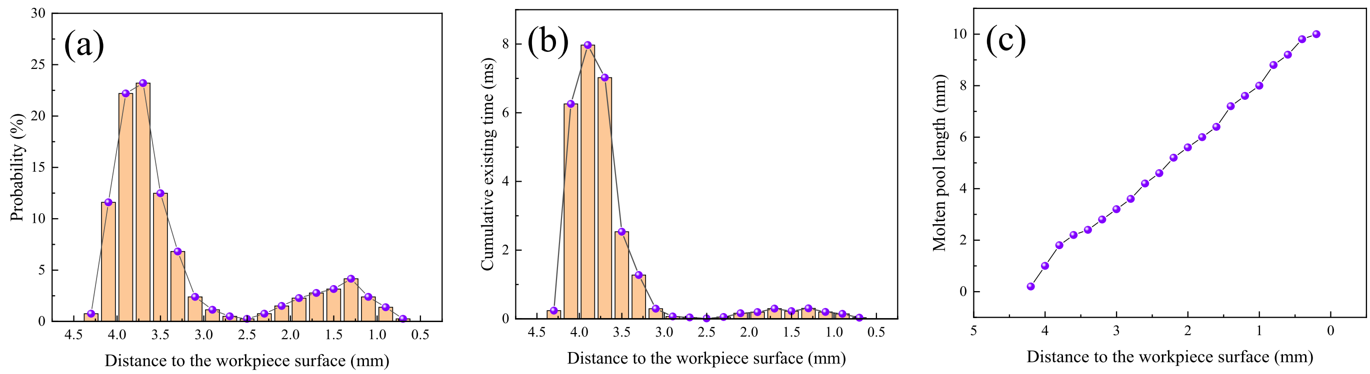

Fig. 5. Statistically analyzed or extracted data along the thickness direction from the numerical model: (a) probability of keyhole collapses, (b) cumulative existing time of collapses, (c) molten pool length.

3.1. Dataset construction

Extensive experimental and numerical investigations have shown that a stable keyhole can hardly be generated from an ideal force balance on the keyhole wall [30,31]. The keyhole experiences transient fluctuations and collapses at different positions, and the geometry evolves on a microsecond time scale [5]. When the collapse happens at the keyhole tip, the gas inside the keyhole is trapped by the surrounding liquid metal and migrates into the molten pool to form a bubble. Some bubbles are eventually captured by the moving solidification front, resulting in pores in the weld [6,32]. Therefore, the key physical information can be categorized into two main groups: keyhole-related variables, which govern bubble generation, and molten pool-related variables, which influence bubble capture. This classification is also supported by findings from many studies [3,5,6,30–32].

Commonly used features, such as keyhole depth measured at a specific time point or averaged over a short time interval, may lack sufficient statistical representativeness. In this study, statistical analysis is conducted on dynamic keyhole profiles for 200 ms after the molten pool reaches a quasi-stable state. This approach allows for the derivation of the probability distribution of collapse positions along the thickness direction. A representative distribution is shown in Fig. 5(a), revealing a bimodal pattern: collapses predominantly occur near the keyhole tip and, to a lesser extent, in the keyhole opening region, while the middle region rarely experiences collapse occurrence. Recent findings also highlight the significance of the collapse duration, defined as the time interval from the initial contact between the front and rear keyhole walls to the disappearance of that contact, as a key index of keyhole stability [33]. Only collapses with extended duration are likely to evolve into gas bubbles within the molten pool, eventually leading to porosity in the solidified weld. Fig. 5(b) presents the cumulative collapse duration along the thickness direction, showing that collapses in the keyhole opening region tend to have a shorter existing time and, therefore, contribute less to bubble formation compared to those near the tip.

Although the full 3D topology of the molten pool can be obtained from the numerical model, incorporating this level of detail would significantly increase the model's structural complexity and result in prohibitively long training times. To reduce computational cost and ensure consistency with the keyhole stability data shown in Fig. 5(a) and Fig. 5(b), the time-averaged molten pool length over 200 ms is presented as a function of thickness position in Fig. 5(c). In this approach, only the longitudinal profile of the molten pool is used as input to the PIML model. Given that porosity formation is primarily influenced by the molten pool's longitudinal characteristics, this simplification is not expected to noticeably compromise the model performance.

Since the data from the numerical model are spatially discretized based on the mesh size (0.2 mm in this study), the extracted physical information varies in the number of data points depending on how many elements are within the range of the penetration depth. To ensure compatibility with the fixed input dimensions required by the neural network architecture, all physical information is resampled using a linear interpolation to a uniform data size. The number of the original data directly extracted from the numerical model is approximately between 20 and 50. According to the preliminary study, 100 interpolation points can achieve a good balance between feature preservation and computational efficiency.

The dataset of the PIML model contains the extracted physical information described above and the four welding parameters as input features, with the experimentally measured porosity ratio serving as the target. The physical estimator takes welding parameters as inputs and is trained to predict physical information obtained from the multi-physics simulations. A unified data split is applied at the level of welding parameter combinations and is consistently used for all ML models (training set: 43 parameters, 46 samples, test set: 20 parameters, 20 samples). One subset of welding parameters, together with all corresponding experimental data and/or simulation outputs, is used for

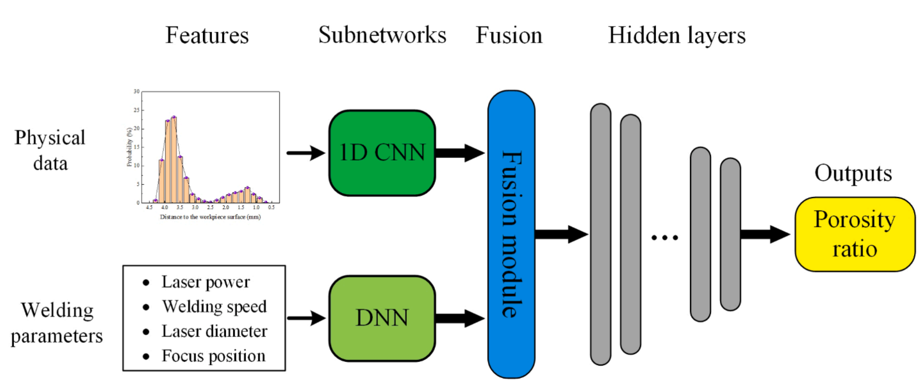

Fig. 6. A schematic overview of the multimodal PIML model.

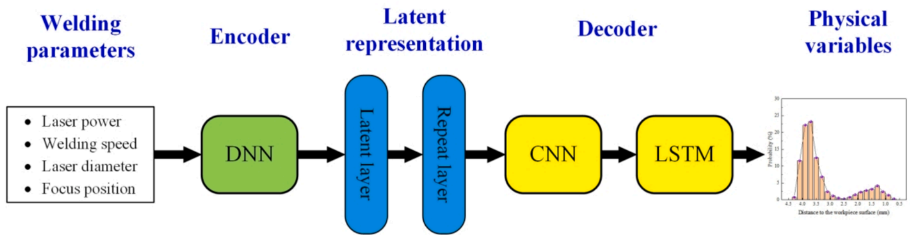

Fig. 7. The architecture of the ML for fast estimation of physical information.

training, while a disjoint subset of welding parameters and their corresponding data is reserved exclusively for testing. For repeated experimental parameters, all corresponding samples are assigned to the same data subset. This parameter-level splitting strategy ensures that no test information is used during training of either model, effectively preventing data leakage. Before training, all input features and targets are normalized to the range [0,1] to mitigate the effects of differing magnitudes.

3.2. Multimodal PIML model

According to the constructed dataset, the proposed model adopts a multimodal architecture that combines a 1D CNN subnetwork to effectively extract spatial features from high-dimensional physical information and a deep dense neural network (DNN) subnetwork to process scalar welding parameters, as shown in Fig. 6. The features from the CNN and DNN subnetworks are integrated into a concatenated layer acting as a shared latent space for a joint representation learning and then passed through additional dense layers to produce the final prediction. The detailed model structure visualized by Keras API Model Plotting Utilities is provided in the supplementary materials. It should be mentioned that this structure can be easily extended with more subnetworks to process multiple data types, such as temperature-dependent material properties, to allow for the porosity prediction with different materials.

Robust fine-tuning of the PIML model's hyperparameters is performed using a grid search cross-validation strategy, which systematically evaluates all possible combinations of predefined hyperparameters. The search space includes the learning rate, kernel size, epochs, and activation functions for both the subnetworks and the output layer. To ensure efficient data utilization, reliable performance estimation, and reduced risk of overfitting, a 5-fold cross-validation is applied to each hyperparameter combination during the grid search. Given the inherent randomness in training deep learning models, such as random weight initialization, prediction results may exhibit variability even when using identical training data and parameters. This variability tends to be more pronounced in complex models such as the multimodal model in this paper. To mitigate this effect, a bootstrap aggregating (bagging) ensemble technique is employed [34]. In this approach, 50 independent sub-models are trained on different bootstrap samples, which are random subsets of the original training data created with replacement. The final prediction is obtained by averaging the outputs of these sub-models, while the standard deviation across predictions is calculated to quantify algorithmic variance.The loss function is defined as mean squared error (MSE), and the model parameters are updated using the Adam optimizer.

The trained models are evaluated based on two performance indexes, the MSE for a general evaluation over the whole training or test set, and the maximum relative error ($err_{\max}$) for the extreme cases, given as:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \widehat{y}_i)^2 \tag{1}$$

$$err_{\max} = \max\left\{\left|\frac{y_i - \widehat{y}_i}{y_i}\right|\right\} \tag{2}$$

where y_i is the true value, $\widehat{y}_i$ is the predicted value, and N is the sample number in the training or test set.

3.3. Fast estimation of critical physical information

Although the multi-physical model can provide a comprehensive and accurate description of the keyhole dynamics and molten pool behavior, its high computational costs and the complexity in performing the

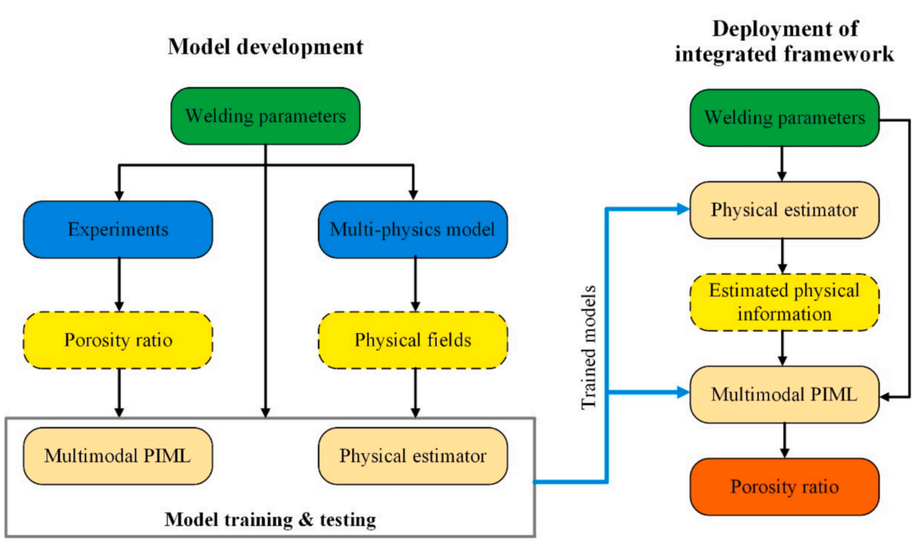

Fig. 8. Data-flow diagram describing relationships among experiments, simulations, the ML models, and the integrated framework.

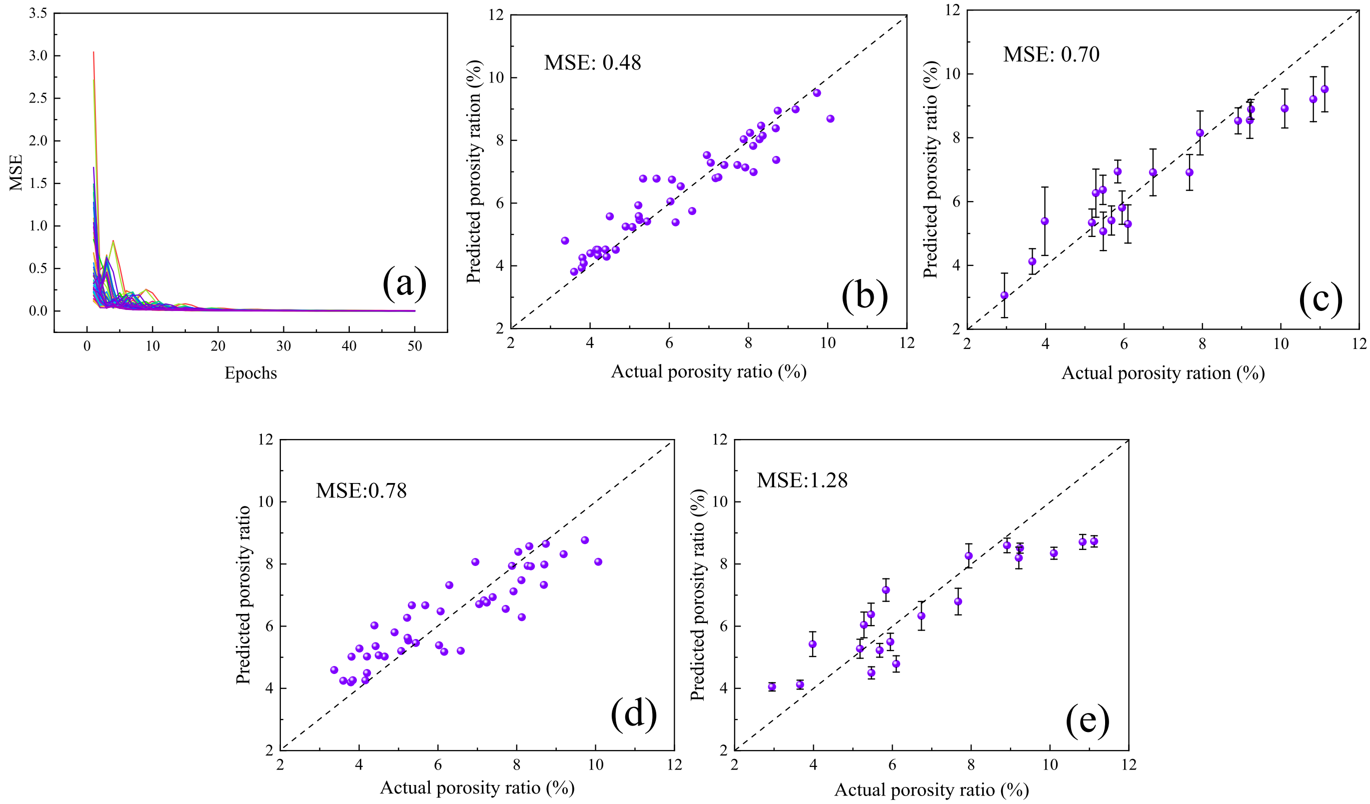

Fig. 9. Comparison of multimodal PIML model with ML model using welding parameter as input: (a) MSE loss function curves of different sub-models from the model ensemble, (b) and (c) are predictions of PIML model for training set and test set, respectively; (d) and (e) are predictions of ML model using welding parameters as input for training set and test set, respectively [23].

statistical assessment make the application of the PIML model difficult. Hereby, an additional ML model will be developed, using the welding parameters as inputs, to realize an estimation of the selected physical information in Fig. 5 in a realistic time frame.

As shown in Fig. 7, this ML model contains an encoder-decoder architecture that transforms a 4-dimensional input (welding parameters) into three generated sequences, each with 100 data points. Initially, the encoder processes the scalar input through DNN layers, expanding the dimensionality, before compressing the information into a latent representation. The latent vector is then repeated 100 times to create a sequence context, which serves as the starting point for the decoder. In the decoder, CNN layers are first implemented to extract local features, and the convolutional output is subsequently processed by LSTM (Long Short-Term Memory) layers, capturing sequential dependencies across the repeated latent features. Finally, a TimeDistributed dense layer is used to generate an output for each of the 100 steps, with each output comprising 3 values, namely the three selected physical information in Fig. 5. The ML estimation model will provide normalized physical information, which can be directly used as inputs of the PIML model. The detailed model structure is provided in the supplementary materials.

The fine-tuning of the hyperparameters follows a similar procedure described in section 3.2. Besides the MSE, the dynamic time warping (DTW) distance is also employed as an index to describe the similarities between two sequences by finding an optimal alignment between them under certain restrictions [35]. A lower DTW distance between the predicted physical information and the ground truth (numerically calculated results) indicates higher similarity or better matching, i.e., a more accurate prediction.

The hardware and software platforms used to develop the two ML models are as follows: Model training is conducted on a workstation equipped with dual Intel Silver 4108 CPUs and an NVIDIA RTX 4080 Super GPU for GPU-accelerated computation. The model is implemented using TensorFlow 2.4 with the Keras API, while grid search cross-validation is performed using Scikit-Learn in conjunction with SciKeras.

Eventually, the whole ML framework (named '*integrated framework*' in the following part) can be constructed by integrating the PIML model and the fast ML estimation model. The physical information predicted by the fast estimation model will be imported as inputs of the PIML model to achieve an accurate prediction of porosity level in a realistic time frame, without performing numerical simulation. Fig. 8 shows a data-flow diagram illustrating the relationships among experiments, simulations, the ML models, and the integrated framework.

4. Results and discussion

The performance of the PIML model and its comparison with the reference case, namely an ML model using only welding parameters as inputs, are presented in Fig. 9. The reference ML model adopts a three-layer DNN architecture that is identical to the DNN subnetwork embedded in the PIML model. The determination of the model architecture and the hyperparameter fine-tuning of the welding parameter-based ML model are described in our recently published work [23]. The multiple MSE loss function curves of the PIML with model ensemble are plotted in Fig. 9(a). All the sub-models exhibit good convergence, with a dramatic drop in the loss function until reaching a stable level. A good agreement (MSE: 0.48) between the actual and predicted porosity level is achieved on the training set, see Fig. 9(b). The trained PIML model shows a good performance in predicting the porosity level on the test set with an MSE of 0.78. It is comparable to the recent PIML work of Ma et al., using the OCT-measured keyhole depth and numerically calculated keyhole temperature as features [22]. Compared with the ML model only trained by welding parameters in Fig. 9(d) and (e), the introduction of physical information improves the prediction accuracy in the overall dataset, with an MSE reduction of 38% with the training

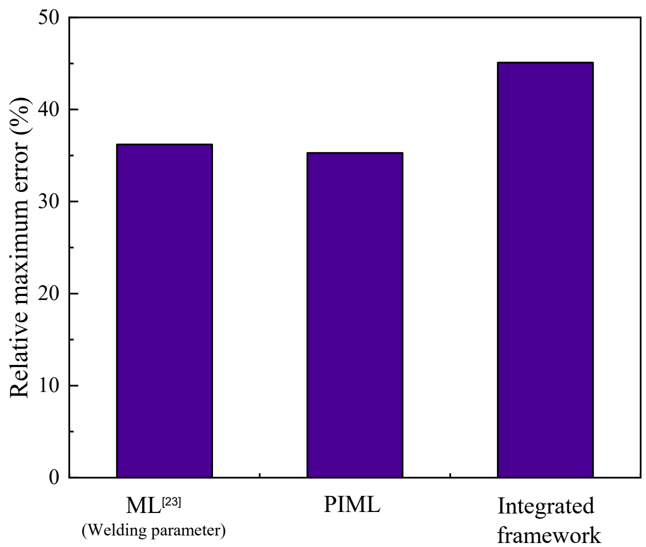

Fig. 10. Relative maximum error of ML model trained only with welding parameters in our previous work [23], the multimodal PIML model, and the integrated framework.

set and of 45% with the test set.

However, no noticeable improvement with the maximum relative error is obtained by the PIML model, as given in Fig. 10. Both models have a maximum relative error of 35% in the low porosity range (below 4%).

Two feature evaluation techniques, namely Shapley Additive Explanation (SHAP) analysis and Permutation Feature Importance (PFI), are employed to reveal the contribution of different features in the porosity formation. The SHAP method is proposed based on the additive feature contribution and the Shapley values from game theory. The idea of the additive feature contribution method is to approximate complex models with a simpler and explainable model to obtain interpretable decisions. The Shapley value describes the significance of an individual feature with its mean marginal contribution, which is defined as the difference between the results produced with/without the objective feature among all possible combinations of features [36], as mathematically given below:

$$\phi_j^i = \sum_{S \subseteq K/\{i\}} |S|!(|K| - |S| - 1)![f_{S\cup\{i\}}(x_{S\cup\{i\}}) - f_S(x_S)]\frac{1}{|K|!} \tag{3}$$

where ϕ_j^i is the Shapley value of the i-th data point in the j-th feature, f is the trained model, K is the set of all features, $S \subseteq K$ is all feature subsets, x_S is the input feature values in subset S, and! represents a factorial operation. In principle, the effect of a feature on the model output is evaluated by computing the difference between one model $f_{S\cup\{i\}}$ with the target feature and another model f_S without the target feature. The overall feature importance can be calculated by averaging the absolute Shapley values per feature across the data:

$$I_j = \frac{1}{n}\sum_{i=1}^{n}\left|\phi_j^i\right| \tag{4}$$

where n is the number of data points of each feature.

The PFI is operated by measuring the increase in prediction error when the values of a single feature are randomly shuffled, thereby breaking the relationship between that feature and the target variable. The original model error (MSE in the present) is represented as e_{orig}. For each feature j, a feature matrix $\boldsymbol{X}_{\text{perm},j}$ is generated by permuting feature j in the original feature matrix $\boldsymbol{X}$. The model error based on the predictions of the permuted data can be expressed as:

$$e_{perm,j} = \frac{1}{n}\sum_{i=1}^{n} L\left(y^i, f\left(X_{perm,j}\right)\right) \tag{5}$$

where y^i is the actual value, L is the error measure. The PFI of the j-th feature is simply defined as:

$$PFI_j = e_{perm,j} - e_{orig} \tag{6}$$

If the model's performance degrades significantly after the permutation, the feature is considered important [37].

It is important to emphasize that the welding parameters and physical information were evaluated and ranked independently due to their distinct physical interpretations, which preclude direct comparison. As presented in Fig. 11, despite employing evaluation methodologies based on differing operational principles, identical ranking sequences emerged for both welding parameters and physical information, indicating the robustness of the evaluation conducted in this study. Specifically, laser power and welding speed are identified as having the most critical influence on porosity formation, followed by laser diameter, while the influence of focus position is comparatively minimal. Among the physical information considered, the molten pool length exhibits the greatest impact on porosity formation, consistent with our recent study [23]. It is also noteworthy that the duration of each keyhole collapse is more influential in determining porosity formation than the probability of collapse. In previous studies, the analysis and discussion usually focused

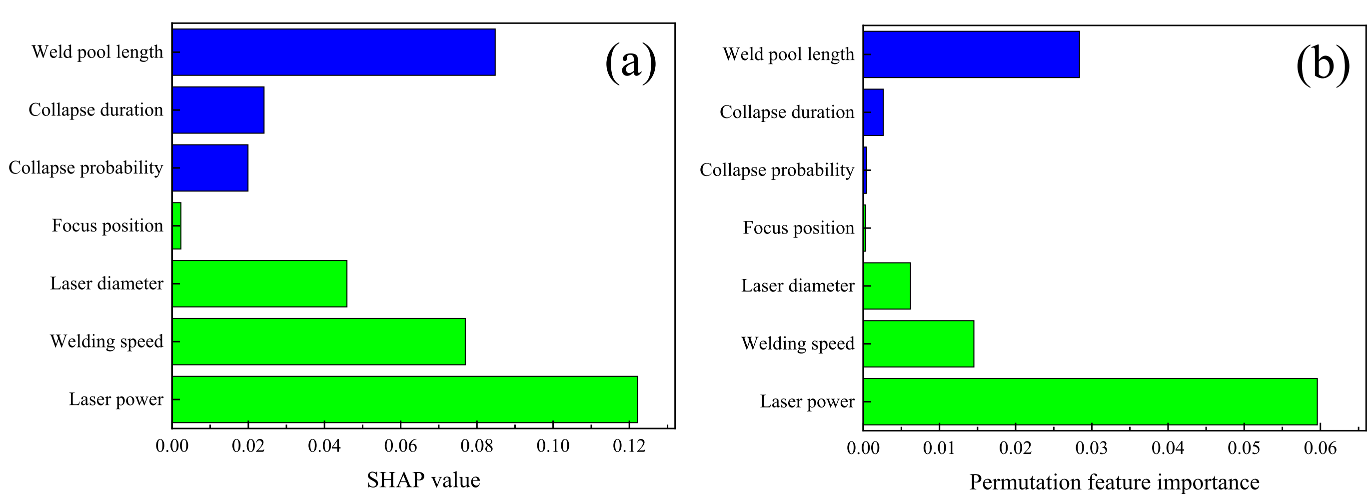

Fig. 11. Contributions of different features to the prediction of porosity ratio by the multimodal PIML model: (a) SHAP value, (b) permutation feature importance.

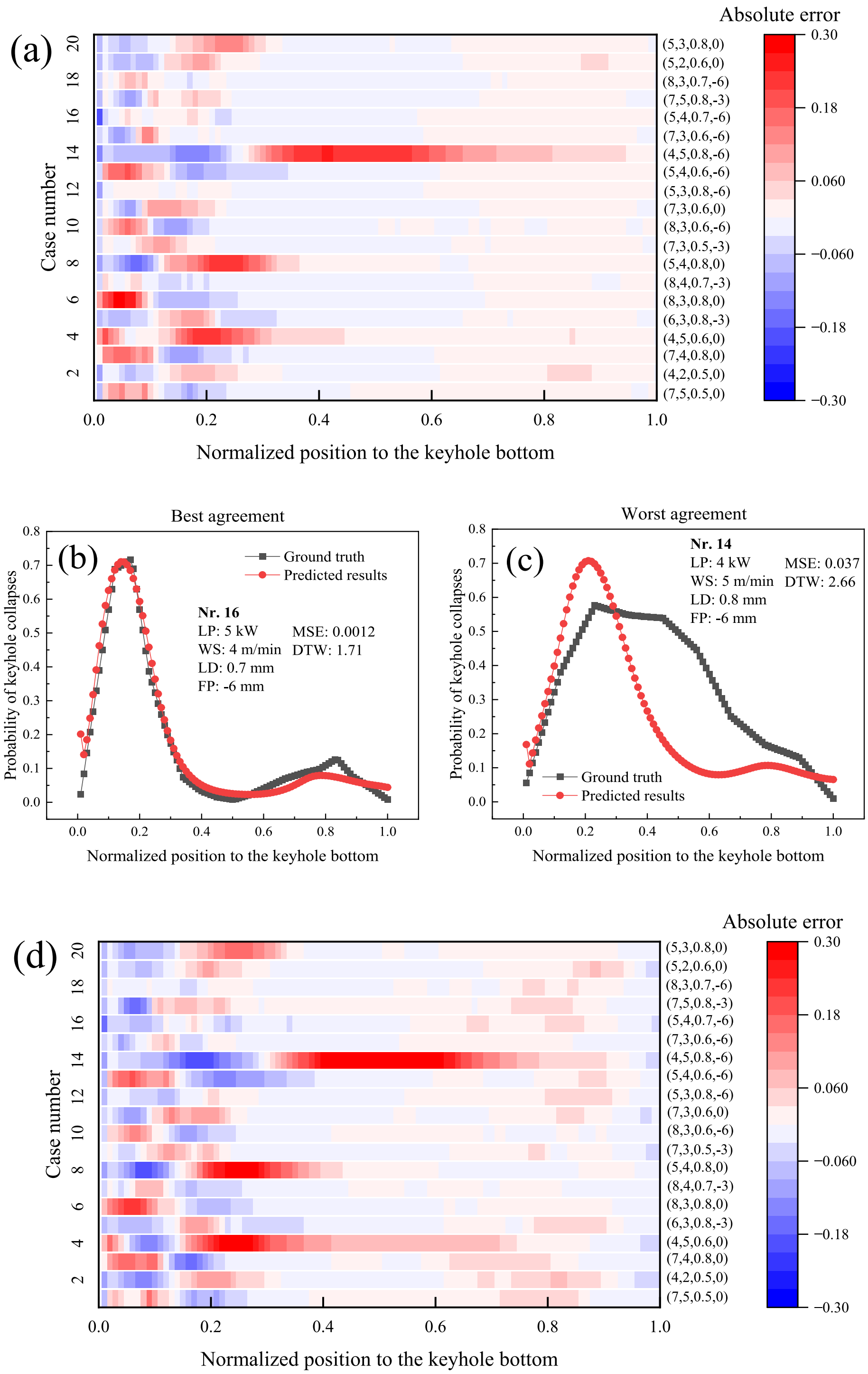

Fig. 12. Error heat map and two representative predictions (best and worst agreements with the ground truth) of the ML model for fast estimation of physical information with test set: (a) – (c) probability of keyhole collapses, (d) – (f) cumulative existing time of collapses, (g) – (i) molten pool length. The ML model predicts normalized physical information, which can be directly fed into the PIML model. LP, WS, LD, and FP represent laser power, welding speed, laser diameter, and focus position, respectively.

on the collapse probability [5,38], and the importance of the duration of keyhole collapse has been underestimated by far.

It should be noticed that the relative differences between physical features appear reduced in the PFI view. This arises from both metric differences: 1. SHAP reflects local prediction-level impact, whereas PFI reflects global performance drop, and 2. correlated inputs such as collapse duration and probability still receive SHAP attribution but have little marginal effect when permuted individually in PFI.

Fig. 12 illustrates the error heatmaps corresponding to the prediction results of the ML estimation model for various physical information, together with the instances representing the best and worst predictive agreement with ground truth data. An exhaustive comparative analysis covering all individual cases is provided in the supplementary materials. As depicted in Fig. 12(b), the ML model reproduces the bimodal feature

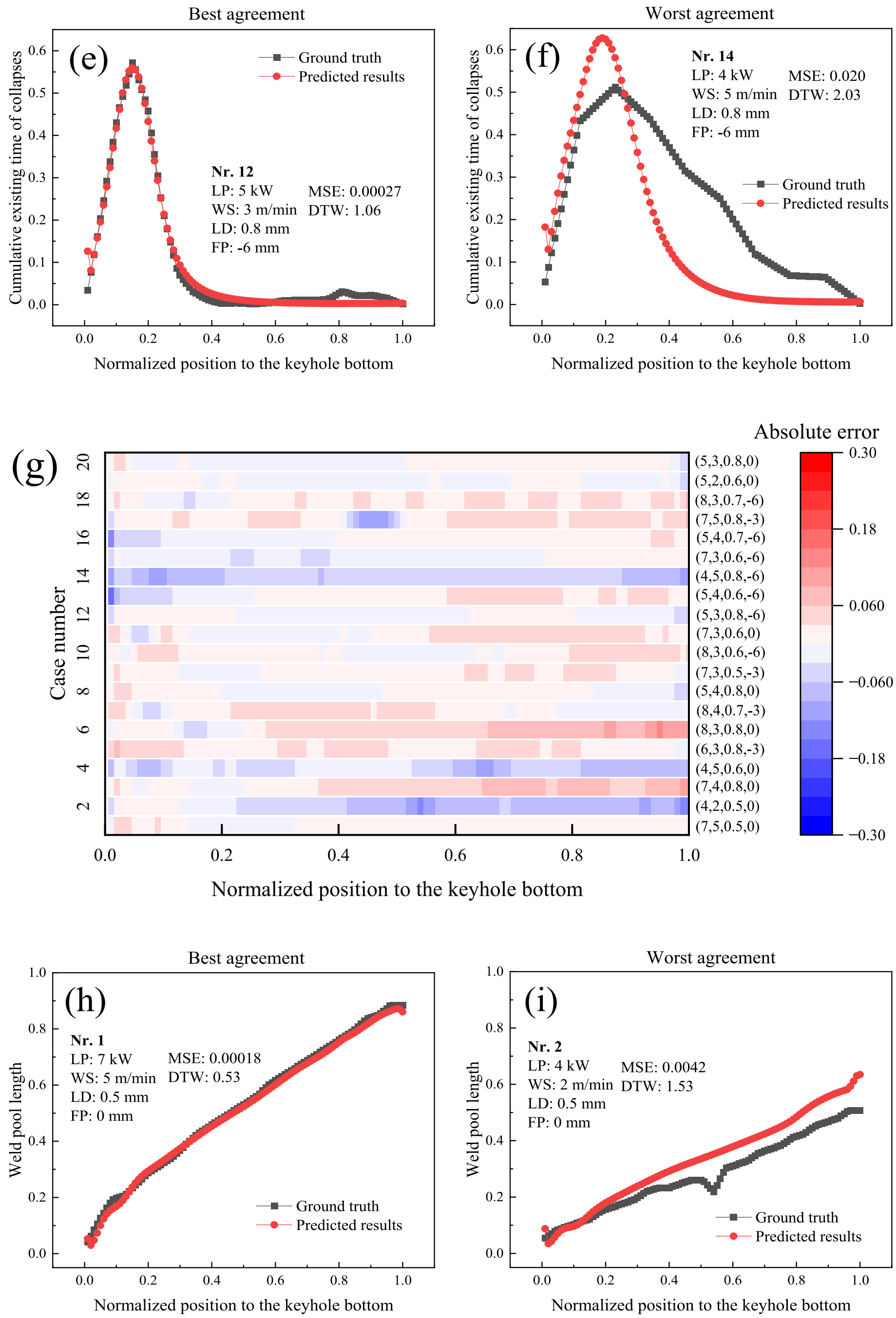

Fig. 12. (*continued*).

in the probability distribution of keyhole collapses, with prediction errors below 10%, reflecting the prediction performance for most test cases. Nevertheless, for the extreme cases, such as No. 14, exhibiting the lowest laser power, highest welding speed, largest laser diameter, and deepest focal position, the predictive capabilities decline notably, resulting in errors as high as 30%. Similar predictive behavior of the model is observed concerning the cumulative existing time of collapses, as illustrated in Fig. 12(d)-(f). It is indicated that the ML model successfully builds a broad correlation between the welding parameters and the keyhole dynamics, which follows a Gaussian or bimodal distribution, but it confronts model-specific limitations, probably due to the architecture or algorithm of the model, with certain outliers that significantly deviate from a Gaussian or a bimodal form.

The molten pool length exhibits an approximately linear variation along the thickness direction across all tested welding parameters. Accordingly, predictions provided by the ML model for molten pool length achieve higher accuracy, maintaining a maximum error of less than 10%, as demonstrated in Fig. 12(g). A nearly perfect alignment between the predicted and ground truth values is observable for case No. 1. Nonetheless, certain cases reveal a minor reduction in molten pool length in the middle region, indicative of bulging phenomena. The predictive accuracy of the ML model in these instances, while slightly lower, remains within an acceptable level (MSE: 0.0042; DTW: 1.53).

With the hardware configuration used in this study, this ML model can realize a fast and accurate estimation of the decisive physical information within 10 s, which is significantly shorter than the calculation time of a CFD model (usually tens of hours). It lays a solid foundation in integrating the PIML model and the ML estimation model into a whole

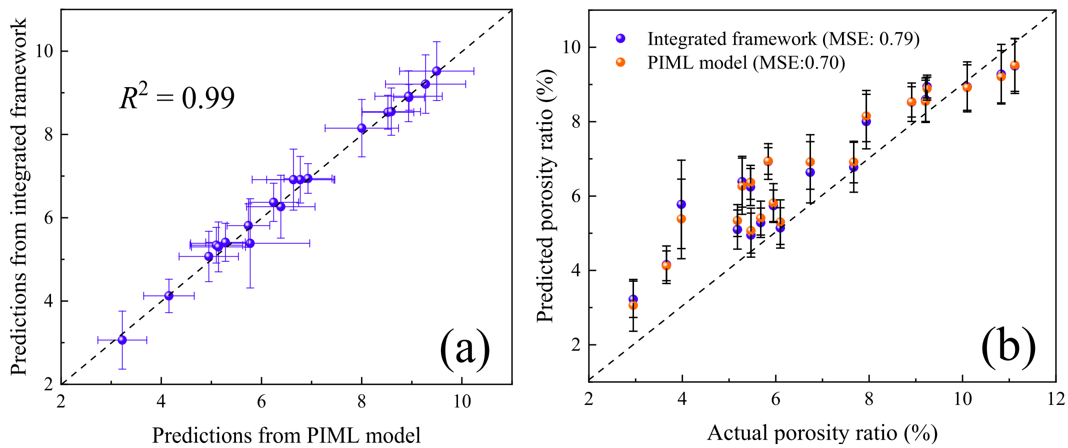

Fig. 13. Comparison of the PIML model and the integrated framework: (a) Correlation between the predicted results, (b) Comparison with the actual porosity ratio.

framework for an effective prediction of the porosity ratio in a realistic time frame.

The predicted physical information is independently provided as inputs to the 50 sub-models developed using the ensemble technique, with the final predictions obtained by averaging the outputs from these individual sub-models. Approximately 40 s are needed to produce a single statistically robust prediction. Fig. 13 presents a comparison of the predictive performance between the PIML model and the integrated framework. As illustrated in Fig. 13(a), predictions from the integrated framework closely align with those obtained from the original multi-modal PIML model, achieving a high R^2 of 0.99. When compared with experimental measurements of porosity ratio, the predictive accuracy experiences only a slight decline, characterized by an increase in MSE from 0.70 to 0.79 and an elevation of the maximum relative error from 35% to 45%, as shown in Fig. 8. The increase in the error is probably caused by the inaccuracy of the predicted physical information from the ML estimation model, especially in the low penetration/low porosity range.

5. Conclusions

With the utilization of multi-physics modeling and experimental data, this paper presents an integrated multimodal physics-informed machine learning framework, achieving an accurate and efficient prediction of the porosity level in LBW of Al alloy. The framework contains two modules: a multimodal PIML model using scalar welding parameters and high-dimensional physical information as input for an accurate prediction of the porosity ratio and an encoder-decoder ML model for a fast estimation of the critical physical information. The main conclusions are summarized as follows:

(1) The multimodal PIML model, which integrates scalar welding parameters and high-dimensional physical information, demonstrates significantly improved predictive accuracy for porosity ratio, achieving a 45% reduction in MSE compared to the ML model trained with welding parameters.
(2) The SHAP analysis and PFI evaluation both reveal that laser power and welding speed exert the most substantial influence on porosity formation, followed by laser diameter, whereas the effect of the focus position is comparatively minor. Among the physical information, molten pool length demonstrates the strongest correlation with porosity formation. The duration of individual keyhole collapses exhibits a greater impact on porosity formation compared to the probability of keyhole collapse.
(3) The developed ML estimation model utilizing an encoder-decoder architecture facilitates a rapid generation of the critical physical information within a practically feasible timeframe (less than 10 s under the hardware configuration employed). The predictions obtained from this model exhibit good agreement with the ground truth across most welding parameter combinations.
(4) The integrated framework of the two ML models can achieve a statistically robust prediction within a timeframe of seconds with only slight deterioration of accuracy.

6. Outlooks

In the current study, all three important physical features are extracted as a 1D ordered sequence (as a function of thickness position) from the numerical model for the simplicity of the data and ML model structure. Therefore, the LSTM technique is employed in the estimator to reproduce the sequential physical features. However, the current model architecture with LSTM cannot be easily extended to 2D or 3D data, whose structure is not explicitly a 1D ordered sequence, for example, a 3D molten pool topology rather than a 1D molten pool length. Further improvements, such as the application of attention mechanisms, should be made for a better extendibility of the model.

Moreover, the model is currently expected to be reliable only within the domain of the training data, and extrapolative predictions should be interpreted with caution. Extra experimental and numerical data beyond the design space are needed to assess the generalizability and robustness of the model. Especially, it should be clarified whether the extendibility of the model is improved by adding the physical features, compared with the ML model relying only on welding parameters.

CRediT authorship contribution statement

Xiangmeng Meng: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Marcel Bachmann:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization. **Fan Yang:** Software, Methodology, Investigation, Data curation. **Michael Rethmeier:** Writing – review & editing, Supervision, Resources, Project administration, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.aei.2026.104611.

Data availability

Data will be made available on request.

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